Problem 15.6

(a) Given $a_{\overline{7}|} = 5.153$, $a_{\overline{11}|} = 7.036$, and $a_{\overline{18}|} = 9.180$. Find i.

(b) You are given that $a_{\overline{n}|} = 10.00$ and $a_{\overline{3n}|} = 24.40$. Determine $a_{\overline{4n}|}$.

Solution

Part (a)

Using the annuity-immediate formula,

$$a_{\overline{m}|} = \frac{1 - v^m}{i}.$$

We are given

$$a_{\overline{7}|} = 5.153, \qquad a_{\overline{11}|} = 7.036, \qquad a_{\overline{18}|} = 9.180.$$

Since $18 = 7 + 11$, we use the decomposition

$$a_{\overline{18}|} = a_{\overline{7}|} + v^7 a_{\overline{11}|}.$$

Substituting the given values,

$$9.180 = 5.153 + v^7(7.036).$$

Solving for v^7,

$$v^7 = \frac{9.180 - 5.153}{7.036}.$$

Thus,

$$v^7 \approx 0.57234224.$$

Taking the seventh root,

$$v \approx 0.9233777.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$i = \frac{1}{v} - 1.$$

Therefore,

$$i \approx \frac{1}{0.9233777} - 1.$$

Finally,

$$i \approx 0.082980.$$

Hence,

$$\boxed{i \approx 8.2980\%}.$$

Part (b)

We are given

$$a_{\overline{n}|} = 10.00 \qquad \text{and} \qquad a_{\overline{3n}|} = 24.40.$$

Using the decomposition of a $3n$-period annuity,

$$a_{\overline{3n}|} = a_{\overline{n}|}\left(1 + v^n + v^{2n}\right).$$

Substituting the given values,

$$24.40 = 10\left(1 + v^n + v^{2n}\right).$$

Dividing by 10,

$$2.440 = 1 + v^n + v^{2n}.$$

Let

$$x = v^n.$$

Then

$$x^2 + x - 1.440 = 0.$$

Solving the quadratic equation,

$$x = \frac{-1 \pm \sqrt{1^2 - 4(1)(-1.440)}}{2}.$$

The positive solution is

$$x = 0.8.$$

Therefore,

$$v^n = 0.8.$$

Now decompose the $4n$-period annuity as

$$a_{\overline{4n}|} = a_{\overline{n}|} + v^n a_{\overline{3n}|}.$$

Substituting the known values,

$$a_{\overline{4n}|} = 10 + 0.8(24.40).$$

Thus,

$$a_{\overline{4n}|} = 29.52.$$